

The Chinese University of Hong Kong, Shenzhen
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CSC5041 Database Systems

Tutorial 1 & 2: Compact Answers

Model answers · Formulas, diagrams and SQL

Adapted from the supplied academic report template.

Original teaching materials remain unchanged.

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Tutorial 1 — Compact Answers

Question sheet, p. 1, Q1(a)–(d). Model answers, not an official marking scheme.

Q1(a) Entities, attributes and keys

Entity	Attributes	Full identifier
Clinic	clinicID, address(city, street, postcode), phone	clinicID
Veterinarian	empID, name, specialisation, salary	empID
Owner	ownerID, name, phone, email	ownerID
Pet (weak)	petNo, name, species, dateOfBirth	(ownerID, petNo)
Treatment (weak)	treatmentNo, date, diagnosis, fee	(ownerID, petNo, treatmentNo)
Medication	medCode, name, unitPrice	medCode

Weak entity	Partial key	Owner entity	Identifying relationship
Pet	petNo	Owner	OWNS
Treatment	treatmentNo	Pet	HAS_TREATMENT

Q1(b) Relationships and ER diagram

`(min,max)` counts counterparts for one entity on that side; `T/P` = total/partial participation.

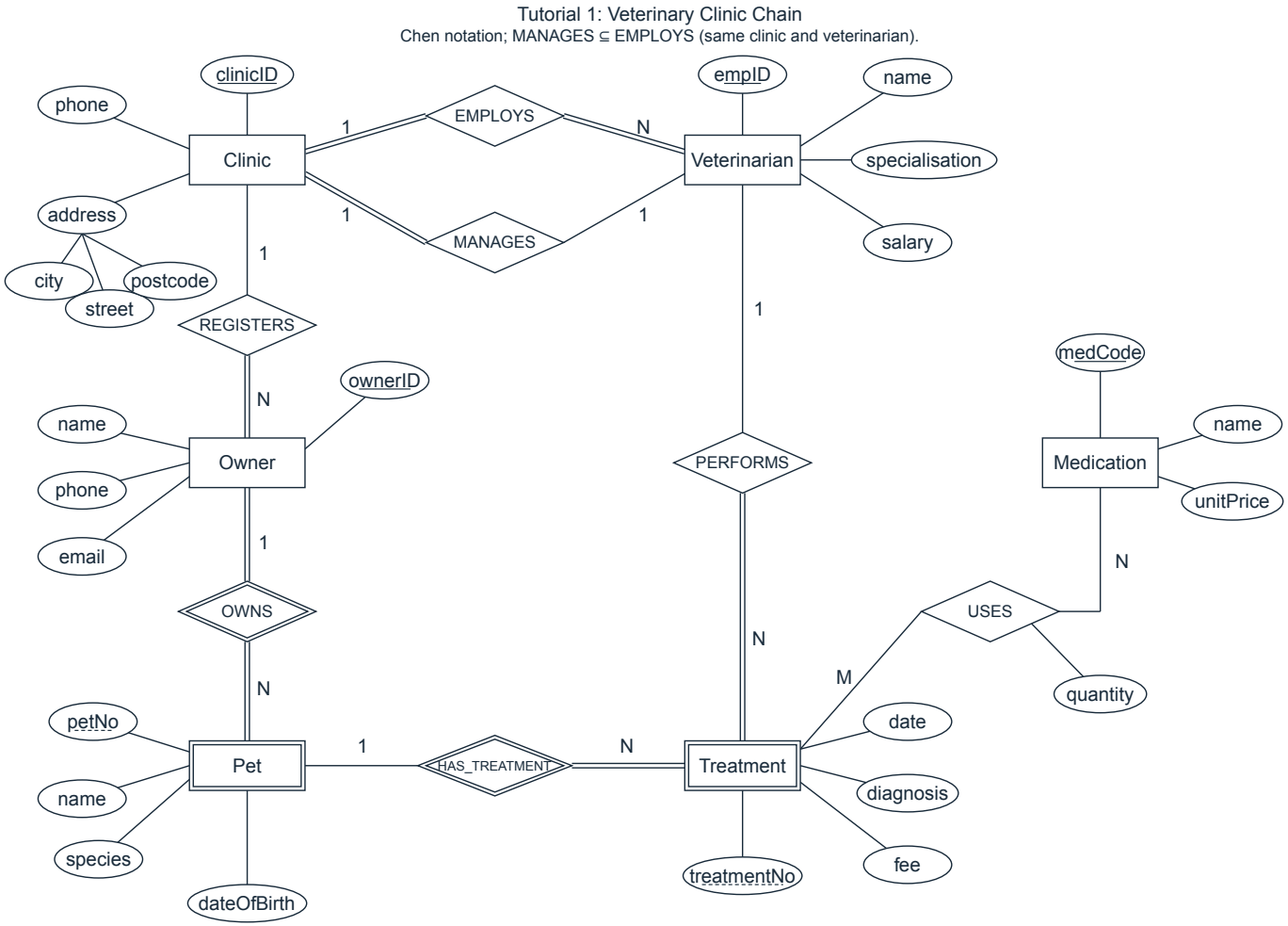
Relationship (left : right)	Ratio	Left	Right
Clinic — EMPLOYS — Veterinarian	1:N	(1,N), T	(1,1), T
Clinic — MANAGES — Veterinarian	1:1	(1,1), T	(0,1), P
Clinic — REGISTERS — Owner	1:N	(0,N), P	(1,1), T
Owner — OWNS — Pet	1:N	(1,N), T	(1,1), T
Pet — HAS_TREATMENT — Treatment	1:N	(0,N), P	(1,1), T
Veterinarian — PERFORMS — Treatment	1:N	(0,N), P	(1,1), T
Treatment — USES — Medication	M:N	(0,N), P	(0,N), P

$\mathsf{MANAGES} \subseteq \mathsf{EMPLOYS}$ [pairs ordered as (clinicID, empID)]

USES has relationship attribute quantity.

Assumptions: unspecified participation is partial; a pet may have zero treatments; one USES row per treatment–medication pair; equal-date treatments are allowed.

Tutorial 1: complete ER diagram



Single/double lines = partial/total; double rectangle/diamond = weak entity/identifying relationship; solid/dashed underline = key/partial key. Clinic's MANAGES link is total; Veterinarian's is partial.

Q1(c) Relational schema

Underlined attributes form each primary key; $*$ marks foreign-key attributes. Composite foreign keys are grouped below.

- Clinic(clinicID, city, street, postcode, phone, managerID*)
- Veterinarian(empID, name, specialisation, salary, clinicID*)
- Owner(ownerID, name, phone, email, clinicID*)
- Pet(ownerID*, petNo, name, species, dateOfBirth)
- Treatment(ownerID*, petNo*, treatmentNo, date, diagnosis, fee, empID*)
- Medication(medCode, name, unitPrice)
- Uses(ownerID*, petNo*, treatmentNo*, medCode*, quantity)

$$\text{Clinic.managerID} \rightarrow \text{Veterinarian.empID}$$

$$\text{Veterinarian.clinicID} \rightarrow \text{Clinic.clinicID}$$

$$\text{Owner.clinicID} \rightarrow \text{Clinic.clinicID}$$

$$\text{Pet.ownerID} \rightarrow \text{Owner.ownerID}$$

$$\text{Treatment.}(\text{ownerID}, \text{petNo}) \rightarrow \text{Pet.}(\text{ownerID}, \text{petNo})$$

$$\text{Treatment.empID} \rightarrow \text{Veterinarian.empID}$$

$$\text{Uses.}(\text{ownerID}, \text{petNo}, \text{treatmentNo}) \rightarrow \text{Treatment.}(\text{ownerID}, \text{petNo}, \text{treatmentNo})$$

$$\text{Uses.medCode} \rightarrow \text{Medication.medCode}$$

$$\text{UNIQUE}(\text{Clinic.managerID})$$

$$\text{NOT NULL}: \text{Clinic.managerID}, \text{Veterinarian.clinicID},$$

$$\text{Owner.clinicID}, \text{Treatment.empID}$$

$$\text{All primary} - \text{key components are NOT NULL.}$$

$$\forall c \in \text{Clinic}:$$

$$\text{Veterinarian}[c.\text{managerID}].\text{clinicID} = c.\text{clinicID}$$

[same-clinic manager: additional cross-relation check]

ER relationship	Representation
EMPLOYS, REGISTERS	1:N → foreign key on N side
MANAGES	1:1 → Clinic.managerID + UNIQUE + NOT NULL
OWNS, HAS_TREATMENT	Identifying → owner key in weak-entity PK/FK
USES	M:N → separate Uses relation; quantity belongs here
PERFORMS	1:N → Treatment.empID

Q1(d) Two constraints beyond keys/foreign keys alone

1. Every owner has at least one pet.

$$\forall o \in \text{Owner}, \exists p \in \text{Pet}: p.\text{ownerID} = o.\text{ownerID}$$

Mechanism: global assertion / transaction-end constraint trigger.

2. Treatment numbers respect chronological order within each pet.

$\forall t, u \in \mathsf{Treatment}:$

$(t.\mathsf{ownerID}, t.\mathsf{petNo}) = (u.\mathsf{ownerID}, u.\mathsf{petNo})$

$\wedge\, t.\mathsf{treatmentNo} < u.\mathsf{treatmentNo}$

$\Longrightarrow t.\mathsf{date} \leq u.\mathsf{date}$

Mechanism: assertion / trigger checking affected histories on INSERT/UPDATE.

Same-clinic management in (c) also needs an extra check; alternatively, declare UNIQUE(empID, clinicID) on Veterinarian and add FK Clinic(managerID,clinicID) → Veterinarian(empID,clinicID).

Source: Tutorial1.pdf p. 1; mapping/notation: Lecture2A.pptx, slides 3, 33–55. Question-sheet SHA-256: `316e8b6bb9581a5ecae3fb2f449e395a4ea8a19b4ee16c4191fc20a6079d2c44`. SQL execution: N/A; schemas/constraints are declarative answers.

Tutorial 2 — Compact Answers

Question sheet, p. 1, Q1(a)–(b). Model answers, not an official marking scheme.

S = Student C = Course E = Enrolled

I = Instructor T = Teaches

$\bowtie$ = natural join; γ = grouping/aggregation; $\leftarrow$ = intermediate relation.

Q1(a) Relational algebra

(i)

$\pi_{\text{title}}(\sigma_{\text{dept}=\texttt{'CS'} \wedge \text{credits}>3}(C))$

(ii)

$\pi_{\text{sname}}(S \bowtie E \bowtie \sigma_{\text{iid}=\texttt{'I1'}}(T))$

$E \bowtie T$ matches both cid and term.

(iii)

$A \leftarrow \pi_{\text{sid}}(E \bowtie \sigma_{\text{dept}=\texttt{'CS'}}(C))$

$B \leftarrow \pi_{\text{sid}}(E \bowtie \sigma_{\text{dept}=\texttt{'Math'}}(C))$

OR : $\pi_{\text{sname}}(S \bowtie (A \cup B))$

BOTH : $\pi_{\text{sname}}(S \bowtie (A \cap B))$

(iv)

$\pi_{\text{sname}}(S \bowtie (\pi_{\text{sid}}(S) - \pi_{\text{sid}}(E)))$

(v)

$D \leftarrow \pi_{\text{cid}}(\sigma_{\text{dept}=\texttt{'CS'}}(C))$

$R \leftarrow \pi_{\text{sid,cid}}(E)$

$U \leftarrow \pi_{\text{sid}}(S)$

$Q \leftarrow U - \pi_{\text{sid}}((U \times D) - R)$

Answer $\leftarrow \pi_{\text{sname}}(S \bowtie Q)$

If D $\neq \varnothing$: $Q = U \cap (R \div D)$. If D = $\varnothing$: `Q = U`.

(vi)

$\gamma_{\text{dept};\,\text{COUNT}(\text{cid})\rightarrow\text{nCourses},\,\text{AVG}(\text{credits})\rightarrow\text{avgCredits}}(C)$

Q1(b) SQL

(i)

```sql
SELECT sname, level
FROM Student
WHERE major IN ('CS', 'Math');
```

(ii)

```sql
SELECT s.sname, c.title, e.term
FROM Enrolled AS e
JOIN Student AS s ON s.sid = e.sid
JOIN Course AS c ON c.cid = e.cid;
```

(iii)

```sql
SELECT DISTINCT c.title
FROM Course AS c
JOIN Teaches AS t ON t.cid = c.cid
WHERE t.iid = 'I1';
```

(iv)

```sql
SELECT DISTINCT sid
FROM Enrolled
WHERE grade IS NULL;
```

`NULL <> 'A'` is UNKNOWN, so WHERE does not retain those rows.

Sources: Tutorial 2.pdf p. 1; Relational_Algebra.pptx, slides 5–59; SQLpart1.pptx, slides 13, 31–36, 44–58. Question-sheet SHA-256: `72c016526c1c0ff02c10fcb5f7f964c1184999c684228671025800936a6cabac`. Expressions/queries are statically derived; SQL has not been executed.